

COURSE DESCRIPTION			
Program	Chemical Engineering		
Course Title	CAD Lab		
Course Code	2079	Semester	II
Type of Course	Lab	Contact Hours	45
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5
CIE Marks	60	ESE Marks	40

Introduction

This course provides hands-on training in computer-aided design, which helps students to develop skills in 2D drafting, isometric drawing, and basic 3D modelling. Emphasis is given to creating accurate engineering drawings by following standard drawing practices, enabling students to effectively communicate design intent and support activities such as fabrication, installation, and maintenance in industrial environments.

Course Objectives

1	To provide hands-on experience in Computer-Aided Design and Drafting.
2	To develop proficiency in 2D drafting and isometric drawings.
3	To apply CAD tools for engineering components and assemblies.
4	To prepare industry-ready drawings following standard engineering conventions.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Engineering Graphics	1003	Engineering Graphics	I
Computing fundamentals	1008	Foundational IT Skills	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Summarise Computer-Aided Design and Drafting.	Understand
CO2	Illustrate the basic commands in AutoCAD	Apply
CO3	Sketch block and Isometric 2D drawing in AutoCAD	Apply
CO4	Illustrate 3D modelling in AutoCAD	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	3	2	-	2	2	3	3
CO2	-	-	-	-	3	2	-	2	2	3	3
CO3	-	-	-	-	3	2	-	2	2	3	3
CO4	-	-	-	-	3	2	-	2	2	3	3
Total	-	-	-	-	12	8	-	8	8	12	12
Correlation Level	-	-	-	-	3	2	-	2	2	3	3

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	0	3	0	1.5	45	1.5				60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Introduction to AutoCAD	History - Applications - Advantages - List commonly used CAD software in industry - Familiarise with the AutoCAD drawing editor.	CO1	Understand	3
Module II					
2	Initiating an AutoCAD drawing	Initiate an AutoCAD drawing. Setting the paper size, space, setting Limits, Units, Zoom, Use of Snap - Grid - Axis - Function keys and Utility commands.	CO2	Apply	3
3	Drawing commands	Illustrate drawing commands: - LINE - POLYLINE - CIRCLE, ARC - RECTANGLE - POLYGON. Apply coordinate input methods (absolute, relative, polar) - Use OBJECT SNAP and GRID settings - TEXT - HATCH - LAYERS - Line types - Selection of objects.	CO2	Apply	3
4	Modify commands and dimensioning commands.	Illustrate, modify, and dimensioning commands: ARRAY - OFFSET - MIRROR - MOVE - ROTATE - TRIM - EXTENT - JOIN - BREAK - CHAMFER - FILLET.	CO2	Apply	3
5	Shaft Bearing	Draw the sectional view of the shaft bearing with dimensions using basic commands.	CO2	Apply	3
6	Square nut and bolt	Draw the sectional view of a square nut and bolt with dimensions.	CO2	Apply	3
7	Hexagonal nut and bolt	Draw the sectional view of a hexagonal nut and bolt with dimensions.	CO2	Apply	3
	Series Exam - I		CO2	Apply	3
Module III					
8	Block	Sketch Block and Title Block.	CO3	Apply	3
9	Isometric drawing	Sketch the Isometric drawing of a given object.	CO3	Apply	3
Module IV					
10	Introduction to 3D modelling	Summarise the setting up of 3D drawing in AutoCAD and viewport settings.	CO4	Understand	3
11	3D Commands	Illustrate 3D commands: EXTRUDE, PRESSPULL, REVOLVE, and 3D FILLET.	CO4	Apply	3
12	3D drawing	Sketch the 3D drawing of a given object.	CO4	Apply	3
	Open Ended Experiment		CO4	Apply	3
	Series Exam - II		CO1	Remember	

Open-ended experiment

Objective:

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:

Modify or extend an existing experiment to explore additional parameters or behaviours.
Integrate multiple concepts from the syllabus into a combined application.
Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.
Perform a comparative study or performance analysis of a circuit/system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		

1	Identify the uses, unique feature and real industrial example of any industry-specific CAD software. Pre-lab. Interpreting the theory concepts and finishing the lab record.	1.5
Module II		
2	Practice an AutoCAD drawing. Completion of the lab record.	1.5
3	Practice drawing commands with the help of simple diagrams. Completion of the lab record.	1.5
4	Practice modifying and dimensioning commands with the help of simple diagrams. Completion of the lab record.	1.5
5	Attempt MCQs and short answers related to CAD fundamentals from the resources. Pre-lab. Interpreting the theory concepts and finishing the lab record.	1.5
6	Learn ASME B 18.2.1 standards for Square nuts and bolts. Pre-lab. Interpreting the theory concepts and finishing the lab record.	1.5
7	Learn ASME B18.2.1 standards for Hexagonal nuts and bolts. Pre-lab. Interpreting the theory concepts and finishing the lab record	1.5
8	Preparation for Series Exam I	1.5
Module III		
9	Learn isometric drawing principles from reference books or videos. Pre-lab. Interpreting the theory concepts and finishing the lab record	1.5
10	Learn to identify front, top and side views of an isometric view. Interpreting the theory concepts and finishing the lab record.	1.5
Module IV		
11	Practice 3D drawing in AutoCAD. Finishing the lab record.	1.5
12	Convert one previously drawn 2D object into 3D. Completing the lab record.	1.5
13	Design a standard title block including name, roll no, scale, and date. Completing the lab record.	1.5
14	Preparation for open-ended experiment.	1.5
15	Preparation for Series Exam II	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Engineering Graphics for diploma	PI Varghese	VIP Publishers

REFERENCE			
Sl. No.	Title	Author	Publication
1	Advanced AutoCAD 2013 - Exercise workbook for Windows.	Cheryl R.Shrock	New Age International Publishers
2	Engineering Drawing and Graphics + AutoCAD	K. Venugopal	New Age International Publishers
3	CAD/CAM/CIM	P.Radhakrishnan, Subramanyan, V. Raju	New Age International Publishers

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://knowledge.autodesk.com/support/autocad/learn	I,II & III

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
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1	Desktop computer systems	Intel i5/Ryzen 5 or higher, 8GB RAM, 512GB SSD	30
2	Licensed operating system	Windows 10/Windows 11 (64-bit)	30
3	Printer	A4 / Laserjet	1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	AutoCAD Software	Latest version/Educational license (2D/3D features enabled)	30

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to		30	15	15	10	
Marks	5	30	15		10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10

Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50

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